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Digital Signal Intelligence

A New Information Substrate for Insurance

Abstract:

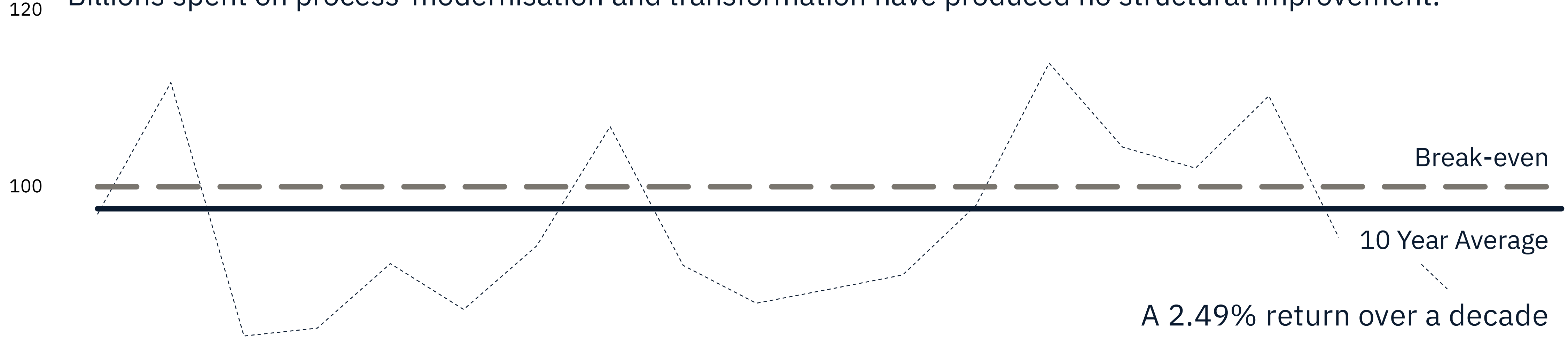
Digital Signal Intelligence (DSI) defines a new, externally observable information substrate for insurance, enabling deterministic, autonomous risk assessment, exposure estimation, loss propensity inference, and pricing at scale.

The Automation Myth

Decades of Poor Returns

Twenty years of Lloyd's combined-ratios show the same truth as every major market: insurance is flat-lining.

Billions spent on process-modernisation and transformation have produced no structural improvement.



Source: [Archived Graphs | III] | [Financial results archive - Lloyd's]

The core assumption is that traditional underwriting inputs can be automated.

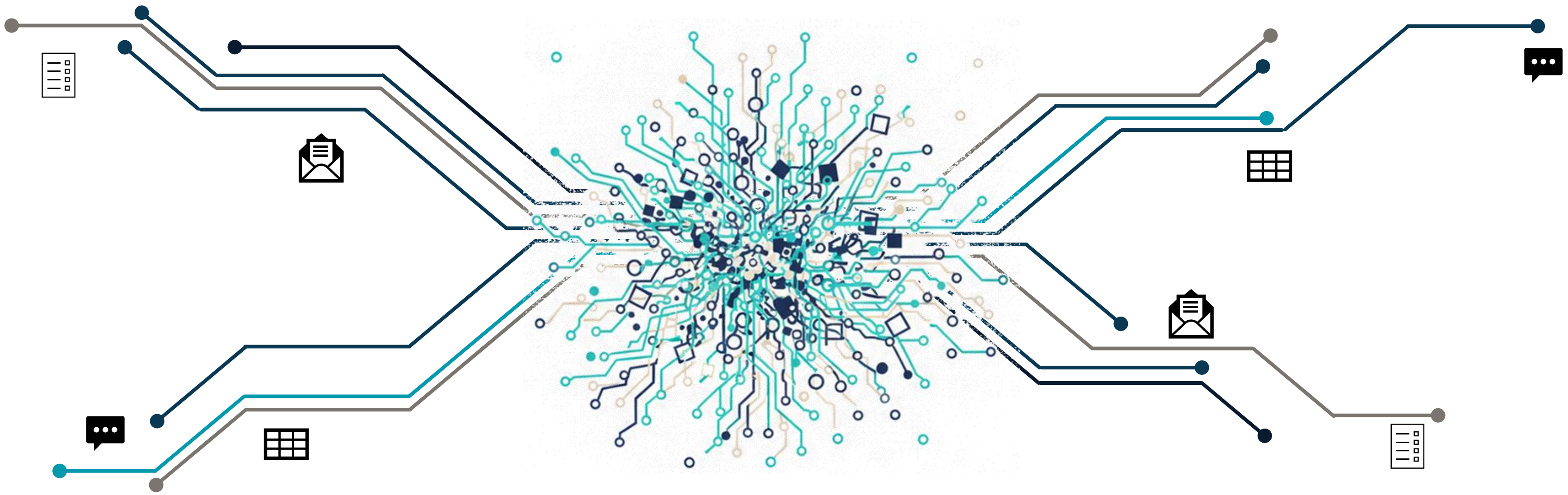
This assumption is incorrect.

The inputs themselves are the constraint.

The Automation Myth

The Information Substrate Problem

Traditional Insurance inputs exist in infinite formats with no canonical schema nor deterministic structure.



No extraction system, including modern or future AI models, can consistently transform these into the inputs required by zero-touch underwriting.

A system built on these inputs cannot achieve deterministic automation.

Autonomous underwriting requires the information substrate to change.

Digital Signal Intelligence

The Core Thesis

Insurance underwriting requires understanding two properties of an organisation:

1. How well it is managed
2. How it behaves over time

Traditional underwriting attempts to infer these properties from self-reported documents, which are subjective, inconsistent, and non-verifiable.

DSI replaces them with observable digital signals.

All organisations operate within a digital ecosystem. Their participation in networks of clients, suppliers, partners, regulators, and certification bodies creates a citation network revealing trust and standing. Their digital trail of operational discipline provides a reliable proxy for maturity. Their hiring activity and investor communications reveal financial performance. Even the absence of expected signals is meaningful because well-run companies are predictable and uniform.

DSI uses these observable signals to create a deterministic, machine-readable foundation for underwriting without requiring self-reported input.

Digital Signal Intelligence

Why Now?

DSI is not experimental.

It uses the underlying fabric of the modern web as introduced by Google's PageRank in 1998.

1

The Problem

Search engines ranked pages by counting self-reported keywords, resulting in spam.

2

The Breakthrough

PageRank introduced a new information substrate: quality inferred from citation.

3

The Parallel (DSI)

DSI applies the same principle. Risk quality is inferred from observable digital signals.

This fact, coupled with modern cloud and AI technologies provide ideal conditions for success.

The question is if this is feasible. The question is why insurance has taken so long to do so.

Digital Signal Intelligence

Foundational Principles

DSI operates consistently and autonomously. To ensure this, ten mandatory principles define the methodology:

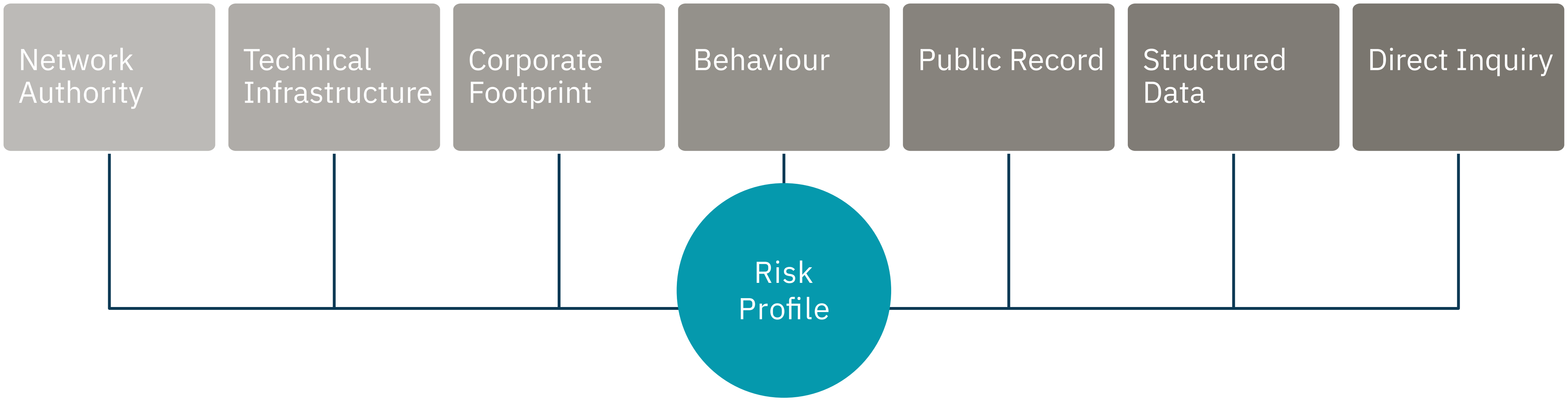
1. External Observability.
   Primary signals must be obtainable without cooperation.
2. Machine Readability.
   All signals must be analysable by automated systems without human interpretation.
3. Network Authority.
   Relationship citation patterns reveal quality
4. Behavioural Inference.
   Observable behaviour is weighted more heavily than stated intentions.
5. Absence as a Signal.
   Absence of expected signals is a measurable indicator of risk.
6. Structured Data Utilisation.
   Authoritative structured data, credit ratings etc, are incorporated directly.
7. Minimal Direct Inquiry.
   Direct questions are optional and constrained to 5-10 per model.
8. Organisational-Level Assessment.
   Risk is assessed at the organisational level; parent behaviour is a proxy for underlying assets.
9. Simplicity in Scoring.
   Models follow a clear scoring flow that is captured and retained at the atomic grain.
10. Agentic Readiness.
    Models must be executable by autonomous systems.

Ensuring consistency, objectivity, and agentic readiness in every assessment.

Signal Architecture

Canonical Categorisation

DSI extracts signals from seven canonical categories that describe how an organisation behaves, operates, and participates in the digital ecosystem.



Signals are externally observable, machine-readable, and verifiable — enabling deterministic scoring without self-reported input.

End-to-End Model Architecture

Structured Workflow

Models convert raw signals into deterministic underwriting outputs through a structured workflow comprising six phases and fourteen discrete steps.

Critically, a post-bind signal loop will transform insurance from a static contract into a dynamic service.

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Three Layer Assessment

Decoupling Complexity for Absolute Clarity

Where traditional underwriting conflates three distinct analytical assessments into a single, subjective, aggregated assessment, DSI separates these into independent, measurable analytical dimensions.

Risk
Utilising a PageRank-inspired digital signal citation framework to observe maturity and predictability.

Exposure
A deterministic inference of an organisation's exposure to calculate an exposure band and complexity score.

Loss
Correlating real-time signal behaviours with historical loss patterns to predict frequency and severity without bias.

Each dimension is assessed independently using the same underlying signal infrastructure, then combined into a unified pricing decision.

Worked Example

Assessing “TechFlow”

The following example demonstrates DSI's end-to-end assessment of a mid-market technology company seeking cyber coverage.

It’s shown using the same end-to-end architecture previously outlined.

1

Discovery.

- DSI validates techflow.io as the corporate website (95% confidence) and identifies TechFlow Solutions Inc. as the legal entity through corporate registry matching.

2

Model Instantiation.

- Model version created, inputs verified against minimum viable requirements.

3

Signals.

- Parallel extraction including 35 cyber-specific signals.

4

Three Layer Assessment.

- Composite Score: 782/1000 → Tier 2
- Exposure Band: Medium ($10M-$50M TIV)
- Complexity Score: Moderate
- Loss Propensity: Low (score 28 / 100)
- Severity Propensity: Moderate (42 / 100)

5

Pricing.

- Final Premium ($5M limit): $20,250
  Base Rate: 0.45% of Limit
  Loss Modifier: 0.90
- No Other Modifiers Applied.

6

Decision.

- Decision: AUTO-APPROVE
  (Tier 2, confidence 0.87, no triggers)
- Total processing time: 47 seconds.
- Full audit trail persisted.

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Competitive Differentiation

Why the Traditional Model Fails Where DSI Wins

DSI occupies a fundamentally different position in the underwriting landscape than existing signal-based tools.

Feature	Traditional	Digital Signal Intelligence
Input	Submissions (PDFs, Excel, Word)	Digital Ecosystem
Process	Manual, OCR, NLP	Deterministic Processing
Integrity	High “Manual Noise" ratio	Zero-Touch, Audit-Persistent
Adjustment	Subjective Overrides	Rule-Based Deterministic Modifiers
Scalability	Limited by manual "Stare & Compare"	Unlimited Agentic Readiness

This is a categorical distinction. DSI is not a better data feed—it is a complete underwriting substrate that happens to consume data feeds, including those from existing vendors, as inputs.

We don't make the old process faster; we make the old process obsolete.

While competitors fight for 10% efficiency gains in data entry, we are rebuilding the foundation.

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Economic Impact

Improvements across the board

The following projections are based on model architecture and assume signal availability ≥ 80% and straight-through processing rates ≥ 60%. Realisation timeline is estimated at 18–36 months from full deployment.

Cohort	Metric	Traditional Model	DSI Target
Expense Ratio	Staff Costs	12%	4%
	Office and Infrastructure	8%	2%
	Administration	4%	1%
	Brokerage and Commissions	11%	6%
	Total	35%	13%
Loss Ratio	Loss Signal Correlation		4-8 points
	Continuous Calibration		2-4 points
	Total		6-12 points
Operational Metrics	Cost per submission	\$150–650	< \$10
	Straight-through rate	5–15%	60–80%
	Time to quote	3–10 days	< 60 seconds
	Scalability	Linear (headcount)	Logarithmic (compute)

On a \$500M premium book, these improvements represent \$130M–\$170M in improved underwriting profit.

The Market

A $1.8 Trillion Global Market — Ready for a New Substrate

DSI applies across the entire commercial insurance landscape because every meaningful organisation now leaves a measurable digital footprint.

This creates a universal, cross-coverage opportunity, with seven configured already.

- PI
- FI
- D&O
- Cyber
- Aerospace
- Marine
- Energy

With an AI Builder configured to help rapidly develop other coverages.

DSI is not a product for one line of business — it is a new underwriting substrate for the entire industry.

Digital Signal Intelligence

The Future of Insurance is Agentic

Digital Signal Intelligence replaces the traditional, self-reported information substrate of insurance with externally observable, machine-readable signals. This enables:

- Deterministic scoring without subjective interpretation
- Autonomous underwriting with deterministic guardrails at scale
- Enhanced Portfolio Steering with Signal weighting optimised at the turn of dial
- Exposure estimation without self-reported schedules
- Loss pattern inference from behavioural signals
- Continuous monitoring and early deterioration detection
- Global scalability without linear headcount growth

Insurers are publicly committing to reducing operational complexity and headcount. However, no traditional underwriting model can achieve this at scale. The bottleneck is the information substrate itself.

DSI is a structural replacement for the information substrate of underwriting, providing a viable path to radical operational efficiency and increased profitability.

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